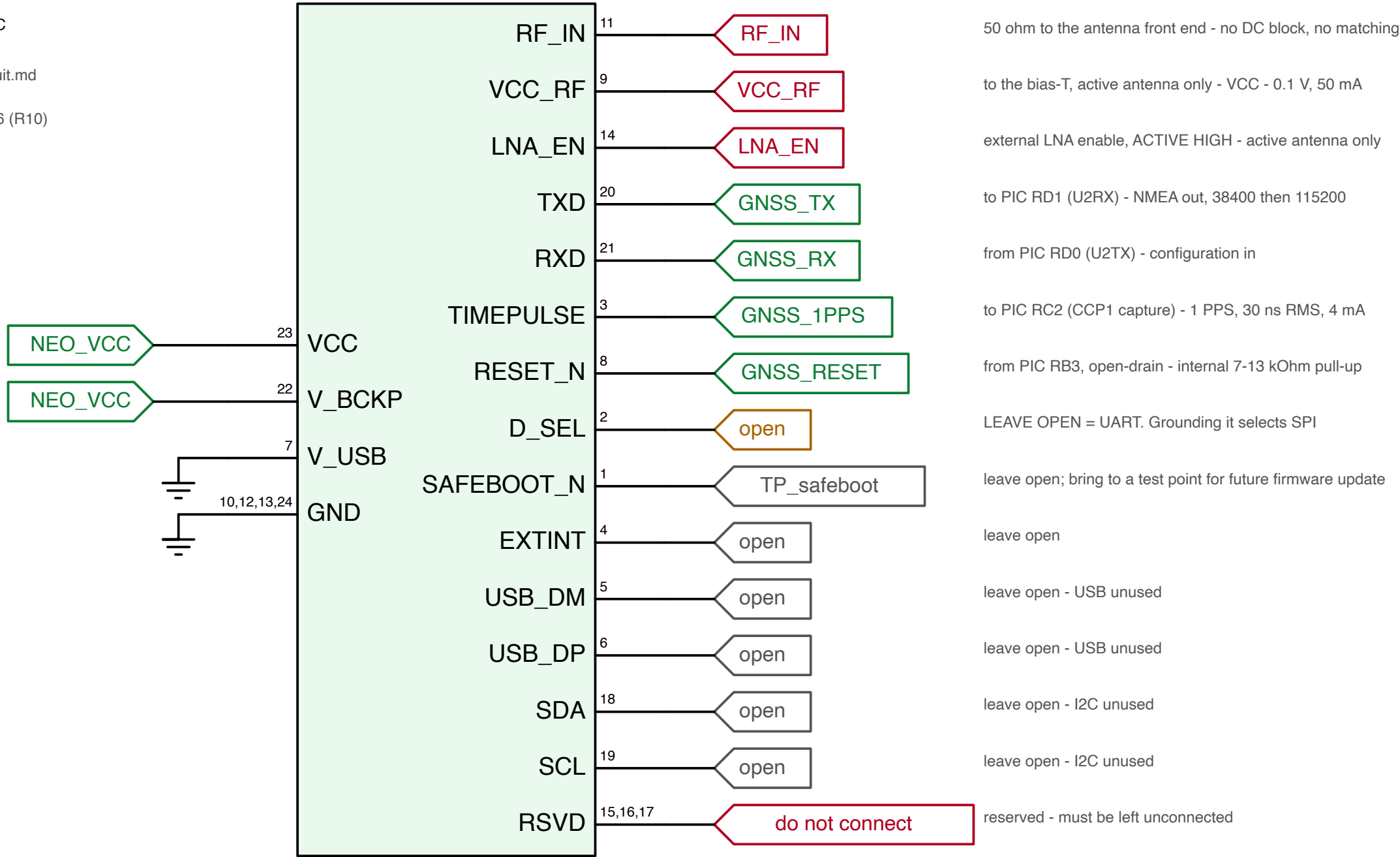


FARSEER GNSS SUPPORT CIRCUIT - u-blox NEO-M9N-00B, 24-pin LCC

Reference designators shown; component values are in gnss-support-circuit.md

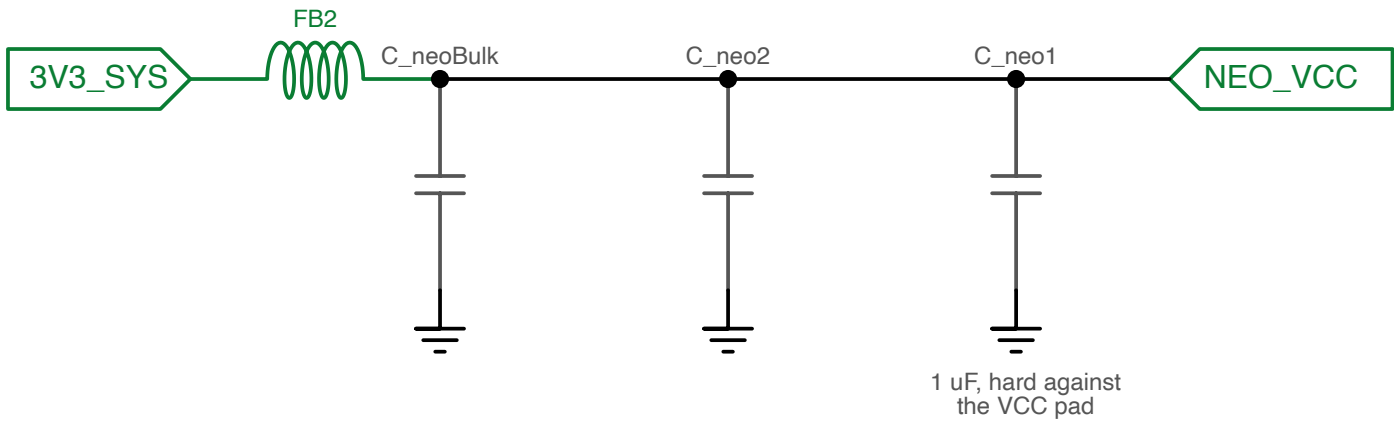
Sources: datasheet UBX-19014285 and integration manual UBX-19014286 (R10)

(1) U4 NEO-M9N-00B : all 24 pins, with the board action for each



Baseline rule (datasheet 5): all inputs have internal pull-ups and can be left open if unused - which is why most of this pin list is "leave open" rather than a resistor. The two exceptions are V_USB, which must be GROUNDED, and V_BCKP, which must be TIED TO VCC if no backup source is fitted. Neither is covered by the leave-open rule. The LCC package has perimeter castellations at 1.1 mm pitch; joints are visible and probeable, no thermal pad, no X-ray needed. Take the footprint and paste mask from IM Figures 37 and 38, and leave keep-out on the long edge for de-paneling residual tabs up to 0.5 mm.

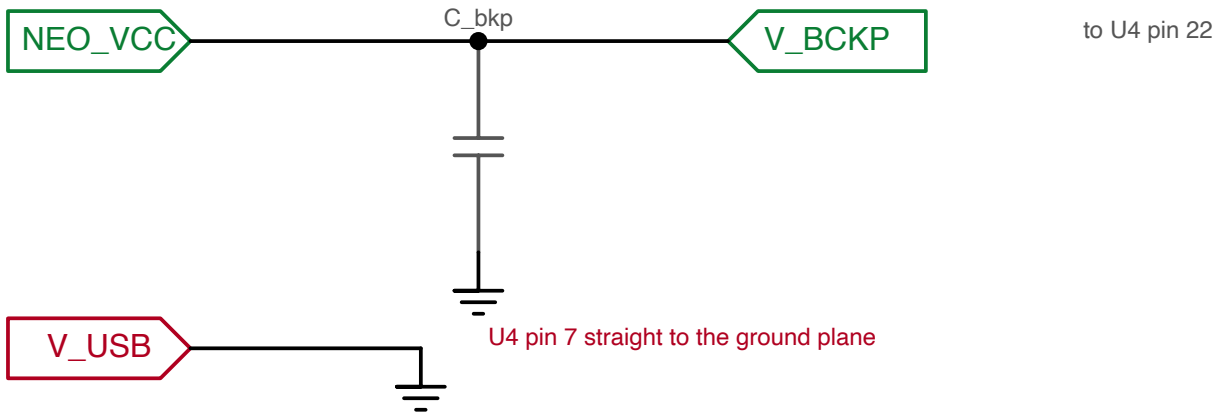
(2) MAIN SUPPLY : 3V3_SYS -> FB2 -> NEO_VCC (pin 23)



THREE CONSTRAINTS THAT ARE EASY TO VIOLATE.

1. Series resistance in the VCC line must stay UNDER 0.2 OHM (IM 4.2.1 / 4.9.1), or dynamic current generates input voltage noise. That is a spec on FB2: choose the bead for DCR below 0.2 ohm, not just for impedance at 100 MHz. A typical 600 ohm @ 100 MHz part can easily exceed it.
2. VCC ramp rate must land between 20 and 8000 us/V, so a 0 -> 3.3 V rise takes between 66 us and 26.4 ms. Too fast MAY PERMANENTLY DAMAGE THE DEVICE (datasheet Table 10 note 7). Buck B's ramp is checked in power-rail-notes.md, and must be re-checked if its soft-start ever changes.
3. The VCC pad wants copper, not a thin trace: large pad, large vias down to the power layer, C_neo1 as close as physically possible (IM 4.8.4.2). Peak draw is 100 mA during acquisition.

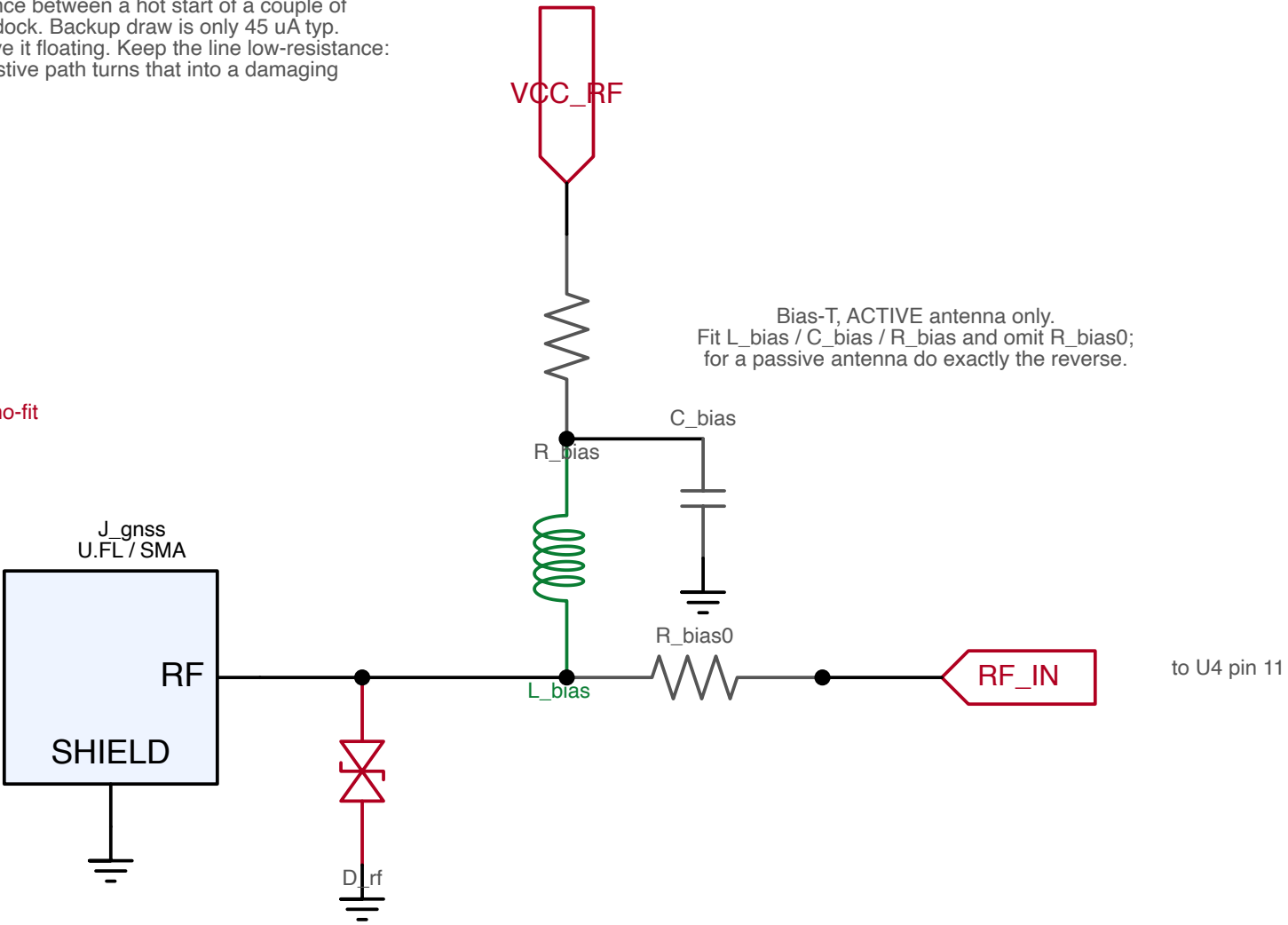
(3) V_BCKP (pin 22) AND V_USB (pin 7)



V_USB MUST BE TIED TO GND when USB is unused (IM 4.2.3, repeated in the 4.3 minimal design and the 4.9.1 checklist). This is not covered by the "inputs may be left open" rule.

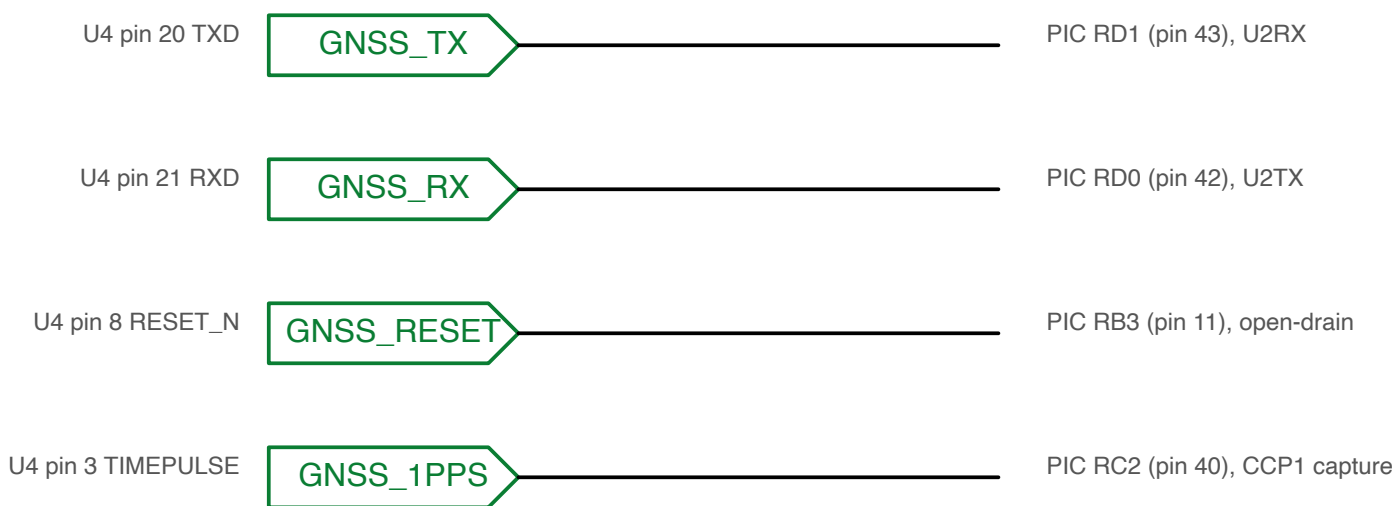
V_BCKP maintains the RTC and battery-backed RAM, which is the difference between a hot start of a couple of seconds and a 24-29 s cold start every time the car is keyed off in the paddock. Backup draw is only 45 uA typ. If no backup source is fitted, TIE V_BCKP DIRECTLY TO VCC - never leave it floating. Keep the line low-resistance: the switchover from main to backup draws a short current peak, and a resistive path turns that into a damaging voltage drop, so no series resistor "for isolation" here.

(4) RF FRONT END - one layout, passive or active antenna by populate / no-fit



The module has an internal DC block and internal 50 ohm matching, plus a SAW filter ahead of its LNA and an LTE band 13 notch. DO NOT FIT A DC BLOCK OR A MATCHING NETWORK on RF_IN - they are already inside, and the SAW is what lets an inexpensive passive antenna work next to a BG95 transmitting at -22 dBm from a few tens of cm away. R_bias is 22 ohm, NOT the 10 ohm in IM Table 26: supplying from VCC_RF at 3.3 V requires >= 19 ohm to hold short-circuit current under 150 mA, and 22 ohm is the nearest standard value. A chated coax against bodywork is the most likely short in a race car, and this resistor is what stops it destroying the bias-T or the internal feed. D_rf is REQUIRED (IM 4.5.2): an external RF connector is a conduction path for destructive signals. RF_IN routes as a grounded co-planar waveguide referenced to layer 2, matched to 50 ohm, via-shielded along its entire length, with vias around the RF_IN pad and NO STUBS. Absolute limits: 0 dBm in band, +13 dBm out of band, 30 dB max external gain - do not stack an active antenna with a further LNA.

(5) UART AND CONTROL TO THE PIC - direct, no level shifting, no series resistors



Both ends sit on 3V3_SYS, so the levels are comfortable in both directions: the NEO needs Vih >= 0.8 x VCC = 2.64 V and swings to within 0.4 V of the rails at 2 mA, and the PIC is on the same rail. Max 5 mA per digital I/O, and NO HARDWARE FLOW CONTROL is supported. TIMEPULSE is worth routing even though the firmware does not use it yet - 30 ns RMS in hardware capture beats UART arrival time by orders of magnitude for correlating position to lap timing.